

# PHYSICS OF COMPLEX SYSTEMS

LECTURE AND TUTORIALS – PROF. DR. HAYE HINRICHSSEN – SS 2026

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Random lecture - Random exercise

## EXERCISE 1.1: RANDOM NUMBERS (6P)

For this programming exercise, you may use `Python` or any other programming language of your choice. Please write and submit your solution in `pdf`-format, including excerpts of the core components of your code, the generated plots and figures, as well as comprehensive answers to the following questions:

- (a) Generate a list of  $N=10000$  integer random numbers between 0 and 9. (1P)
- (b) Determine the relative frequency of the outcomes and plot them in the form of a histogram (a bar chart with 10 bars). (2P)
- (c) On average we expect roughly 1000 outcomes for each digit between 0 and 9, corresponding to a relative frequency of 0.1. What would you roughly expect for the dispersion of the *actual* number of outcomes (that is, the typical size of the random deviation from this average)? (1P)
- (d) Now let us consider the sum of  $K$  uncorrelated random numbers between 0 and 9. Obviously, this sum is an integer random number in the range between 0 and  $9K$ . Which qualitative shape of the histogram would you expect for  $K = 2$ ? After guessing, modify your code accordingly and generate the histogram by yourself. (1P)
- (e) What happens to the histogram for large values of  $K$ ? (1P)

( $\Sigma = 6P$ )

Warmup exercise with only 6 instead of 12 points. Please submit your solution electronically as a single PDF file until Wednesday, April 22, 2026, via WueCampus.

You are welcome and encouraged to use AI-tools for solving these problems. However, please refrain from copy-pasting complete AI-generated solutions.